# Module 1: Introduction to Simulation

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| Book: | Module 1: Introduction to Simulation | | |

# Description

## Introduction to Simulation?

### Prerequisites

- Interest and basic motivation

### Learning Objectives

At the end of this module, you will be able to:

- Define the terms system, model, and simulation and explain them using examples.
- Describe different modeling methods and identify their areas of application.
- Identify continuous and discrete states.
- Differentiate between stochastic and deterministic simulations.

### Time Requirement

- 30-40 minutes

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5. Modeling Methods and Application Areas

6. Completion

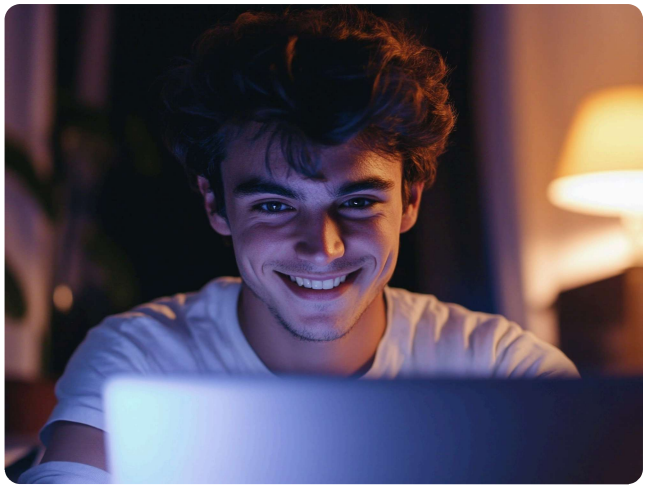
# Welcome to this Digital Learning Unit!

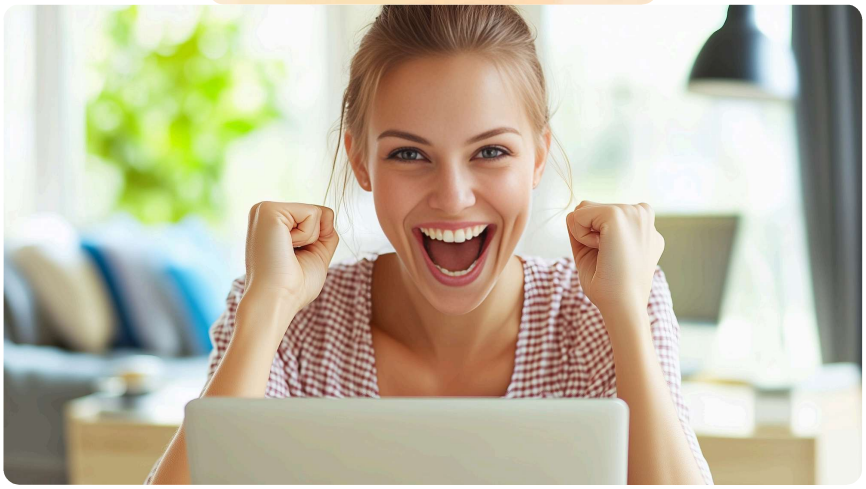
## *Introductory Notes*

The learning unit "What is Simulation?" is a comprehensive digital learning resource designed for self-study. You can learn at your own pace. The course is structured with numerous examples, exercises, and interactive elements to enhance your skills, competencies, and motivation. Your learning progress will be tracked and concluded with a final assessment. Upon successful completion, you will receive a digital e-learning badge as a personal certificate, which will be displayed on the course page.

On the right side of each page, you will find a button to submit your feedback. Please use it! If you encounter any issues or feel that something is missing, let us know!

Also, on the right side, you will find the table of contents and navigation, allowing you to click from one page to the next. This digital learning unit functions like an interactive book that you can read page by page and work through at your own pace.





Enjoy the course and let's get started!

# Introduction to Simulation

## *A look into the diversity and possibilities of modern simulations*

Simulations have become an essential part of everyday life and are used in various fields. A classic example is an aircraft flight simulator, which allows trainee pilots to practice different **flight maneuvers** in a safe environment and prepare for emergency situations. Similarly, **weather simulations** play a crucial role by analyzing meteorological data and creating weather forecasts that are vital for agriculture, transportation, and disaster prevention.

In science, **cosmological simulations** provide insights into the formation and evolution of the universe by modeling complex physical processes. In the economic sector, **logistics simulations** help companies optimize supply chains, efficiently manage inventory, and plan transport routes to reduce costs and increase customer satisfaction.

These examples highlight the broad scope of simulations, from training and education to science and business. They enable us to understand complex systems, minimize risks, and make informed decisions.

In **video 1** you will find a presentation with various examples.

***Watch the video and memorise the various examples and areas of application. The video presentation is in german but there are english subtitles. please make sure they are activated.***

*Video 1:* What is Simulation? by Prof. Dr. Lukas Hollenstein (05:54)

After watching the video, you will be ready for the next chapter!

Every specialism has its own vocabulary. In the next chapter, you will familiarise yourself with the most important terms and their meaning. Contexts are explained and you learn to differentiate between them.

# Terminology: System, Model, Simulation

*From systems to simulations: Modeling as the key to analysis and prediction*

## What is a System?

The term system refers to a set of objects in a defined area that have properties and can be related to each other. If the state of a system can change over time, the system is a dynamic system.

The term ‘system’ in relation to simulation refers to a set of interconnected or interacting components that together have a certain behaviour or function. In a simulation, the system is modelled to investigate and reproduce its dynamics, structure and the interactions of its components.

A system can be a physical, biological, social or technical system, depending on what is to be simulated. For example, a system could be a production plant, an ecosystem, an economic structure or even a transport system.

## What are the characteristics of a system?

Important characteristics of a system in the simulation are

- **Components:** Individual units that make up the system.
- **Interactions:** Relationships and interactions between components.
- **Inputs and Outputs:** The system receives inputs, processes them, and produces outputs.
- **States:** The conditions or properties of the system at a given time.

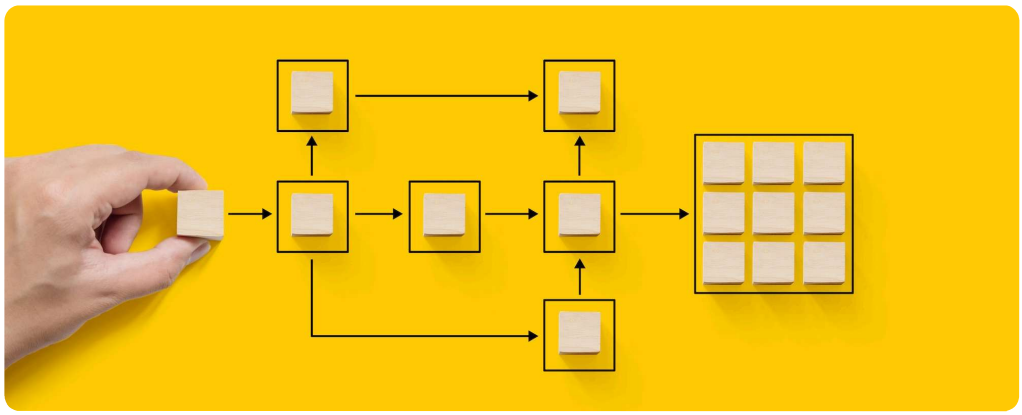


*Figure 5:* The diagram describes a system in which the components are represented by the blocks. Interactions between the blocks take place via connections and arrows that show the flow from inputs to outputs.

Simulations of systems are carried out to analyse their behaviour under different conditions, to support decisions or to make predictions about future states.

## What is a model?

A model is a simplified representation of a system and reflects the temporal and logical-structural behaviour of the system. By experimenting with the model, previously unknown laws of the system can be worked out. A simplified image of the system is usually sufficient for this purpose. It can be mathematical, physical or computerised and serves to capture and understand the essential characteristics of the system. A simulation model is a model of a real system with which, firstly, the internal processes in this system can be analysed, secondly, the effects of interventions in the system and, thirdly, the future course of this system.

## What is a simulation?

Simulation is the recreation of a system with its dynamic processes in an experimental model in order to gain knowledge that can be transferred to reality. Simulations make it possible to make predictions without influencing the real system. Within simulation, a distinction can be made between continuous and discrete simulation. In continuous simulation, the state variable changes continuously, as does the time. In discrete simulation, both the state and the time

change discretely (clocked or from event to event). Hybrid simulations contain continuous and discrete elements.

# General Simulation Approaches

*An overview of modelling methods and simulation types*

Simulation approaches can be categorised according to the following main criteria:

- Change of parameters over time (static or dynamic)
- Type of state transitions (discrete or continuous)
- Change of state variables (time- or event-orientated)
- Describability of the system (deterministic or stochastic)

The classification of simulation models is shown schematically in **Figure 1**. The simulation always has a dynamic component.

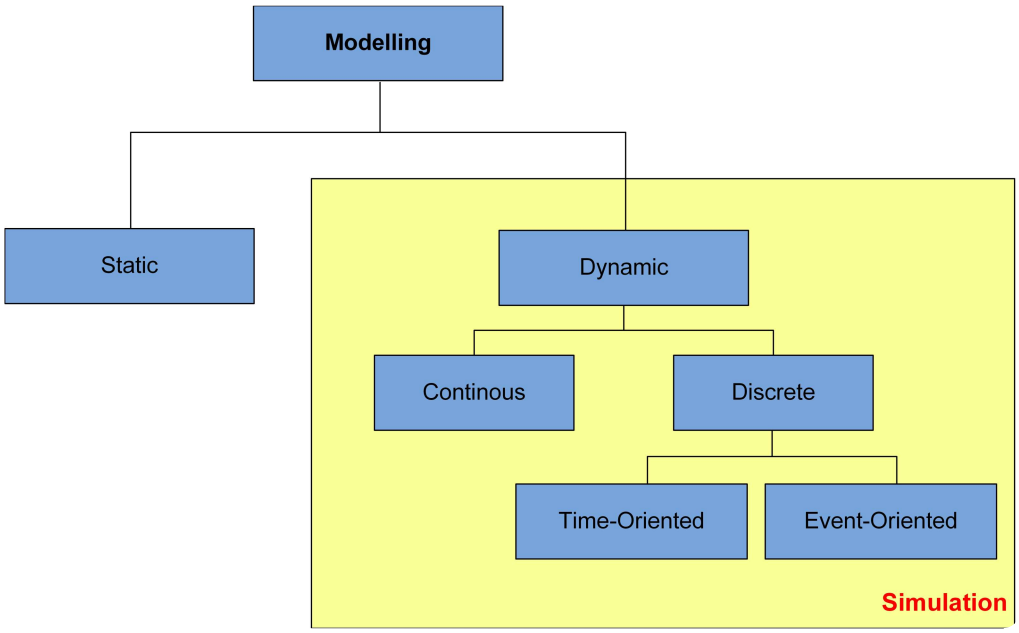



*Figure 1*: Classification of simulation approaches

## Static vs. dynamic

Time plays no role in a static model, as it only considers a fixed point in time and the parameters of the system remain constant over time.

In dynamic models, on the other hand, time is a central factor. The state of the system changes over time. A current state z(t1) is clearly determined by the initial state z(t0) and the temporal progression of the input variables in the period between t0 and t1 (Bossel, 1992).

Dynamic models represent non-stationary processes or temporal changes in stationary processes. The focus here is on analysing and investigating dynamic models and simulations.

## Continuous vs. discrete

In a continuous simulation, the system is described with coupled differential equations, resulting in an apparently continuous flow. However, this is actually based on many small discrete steps, which the human eye perceives as continuous. This simulation approach is characterised by continuous changes over time, both for independent and dependent variables (Cellier, 1991). Continuous-time models are characterised by the fact that the values of their state variables can change infinitely often within a finite period of time.

In contrast, discrete simulation considers a finite number of state changes. The model variables only change at certain, clearly defined points in time (discrete time) and the states are delimited from each other (discrete state). For example, when processing a workpiece on a machine, only the start and end states are simulated; intermediate states during processing are not taken into account (see **Figure 2**).

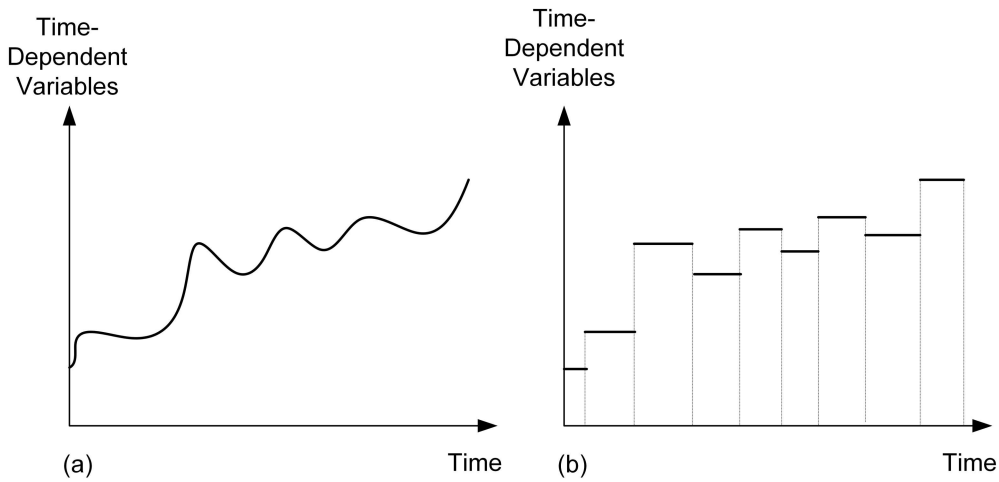



*Figure 2*: Continuous and discrete change of variables

In the case of discrete systems, a distinction is made between time-orientated and event-orientated models.

## Stochastic and deterministic simulations

In stochastic simulations, there are random influences that affect the behaviour of the system. These simulations take uncertainties and

probabilities into account. A typical example is weather forecasting, where it is not possible to predict exactly whether it will rain, but only probabilities for different weather conditions can be specified.

In deterministic simulations, the system follows clear, fixed rules without random influences. This means that the same result is always achieved under the same starting conditions. An example of this would be the simulation of a mechanical pendulum whose movements can be accurately predicted.

In practice, these simulation approaches are often not used in isolation, but in combination. A common example is discrete simulation, which is based on an event-orientated model, also known as discrete event simulation. Here, the states only change as a result of certain events, while they remain unchanged between events.

However, the distinction between deterministic and stochastic simulations is not always clear-cut, as many models contain both deterministic and stochastic elements, as well as static and dynamic components. The dynamics result from the time dependency of the state changes.

*Please watch **video 1** carefully to understand each step and then answer the follow-up questions.*

*Pay particular attention to the video:*

- *the different simulation approaches and their distinguishing features (e.g. static vs. dynamic, discrete vs. continuous).*
- *the explanations of time-orientated and event-orientated models and their respective examples*
- *the differences between deterministic and stochastic simulations and their practical applications*

Eventuell Video-Erklärung oder Abbildung:
https://www.youtube.com/watch?v=7uaQX76e4EI&ab_channel=MikeSaint-Antoine
||640||390

By loading the video, you accept YouTube's privacy policy.

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*Video 1*: Deterministic vs. stochastic modelling by Mike Saint-Antoine (03:23)

Here are some follow-up questions based on the video.

***Follow-up questions to video 1***

What is a typical feature of a static simulation model?

- It uses differential equations for modelling.
- It is based exclusively on random events.
- It analyses a single point in time.
- It takes time changes into account.

## Discrete-time vs. discrete-event model

Discrete-event models are characterised by the fact that the values of the state variables only change a finite number of times within a limited time interval.

Discrete-time models, on the other hand, only change the values of their state variables at predetermined, known points in time.

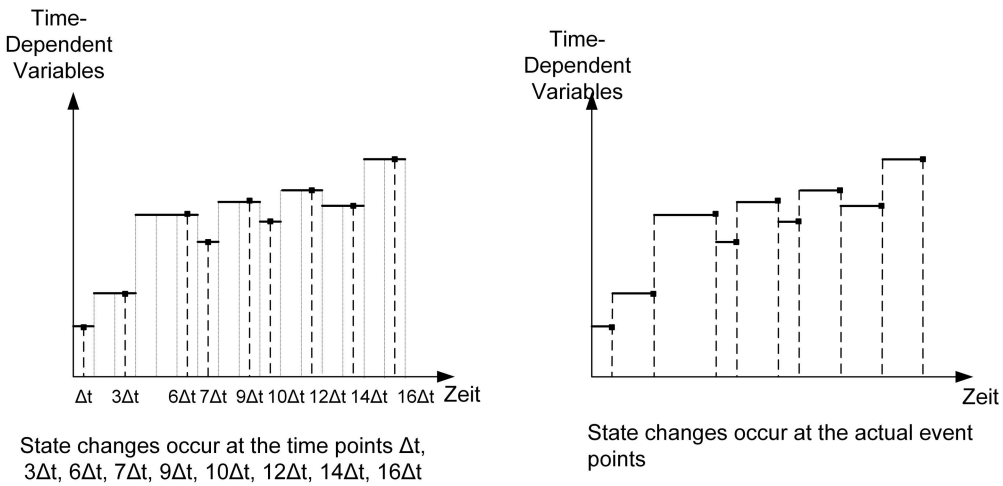


*Figure 3*: Time- and event-orientated simulation

Time-orientated and event-orientated models can be subdivided even further. We will continue in the next chapter.

# Methods and Application Areas

*From time sequences to actors: methods and areas of application of simulation*

## Introduction to the subdivision of discrete models

Discrete models are an essential method for simulating and analysing complex systems. **Figure 4** illustrates the basic subdivision of discrete models into two main categories: time-oriented and event-oriented models (Arnold, Isermann, Kuhn, Tempelmeier, 2004). Time-orientated and event-orientated models can be further subdivided. This distinction is based on the way in which state changes are described in the model.

**The time-oriented models** include approaches such as quasi-continuous simulations, where time serves as the main driver for state changes. They enable the modelling of processes in which changes are observed at regular time intervals.

In contrast, **event-orientated models** are based on specific events that change the state of a system. Within this category, a distinction is made between different approaches:

- **Agent-based models** focus on the interactions of individual actors (agents).
- **Event-driven models** analyse state changes that are triggered by discrete events.
- **Activity-orientated models** track the sequence of activities.
- **Process-oriented models** focus on the modelling of processes and their sequences.
- **Transaction-orientated models** look at the exchange of information or resources between entities.

The illustration thus provides a clear visual classification of the model types and serves as a starting point for understanding the characteristics and areas of application of the various methods.

*Time-orientated and event-orientated models can be further subdivided* (see **Figure 4**)

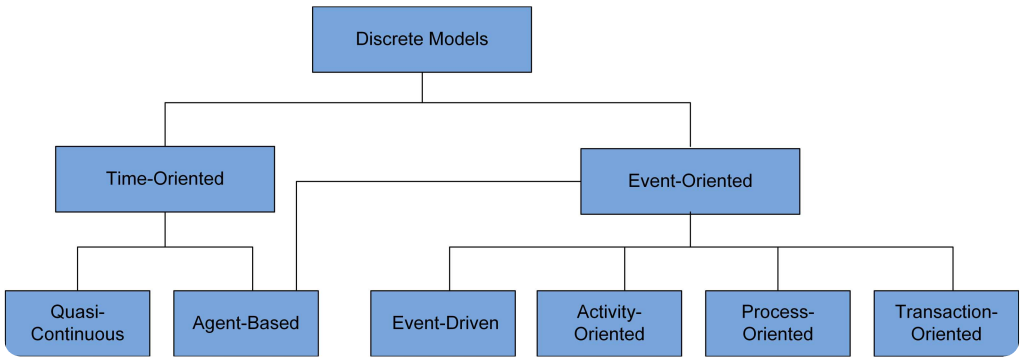


*Figure 4*: Classification of discrete models

The following section presents various modelling methods in detail and describes how they work. The key features of the methods are discussed and typical areas of application in which these models are particularly useful are highlighted.

## 1. Discrete event simulation (DES, event-driven)

Diese Simulationsart beschäftigt sich mit Systemen, bei denen Ereignisse zu diskreten, klar definierten Zeitpunkten stattfinden. Das System wird durch eine Abfolge von Ereignissen verändert, die z.B. das Eintreffen eines neuen Kunden in einer Warteschlange oder das Verlassen einer Maschine nach der Wartung darstellen.

Anwendungsbereiche: Produktionssysteme, Warteschlangensysteme (z.B. im Supermarkt oder bei Telefonhotlines), Logistiksysteme.

## 2. Agentenbasierte Simulation (agentenbasiert)

Here, a system is modelled by the interactions of individual ‘agents’ that act according to certain rules. Each agent (e.g. a person, an animal, a vehicle) has its own characteristics and decision-making processes. These simulations are particularly suitable for modelling complex social or economic systems in which the behaviour of many individual agents influences the overall system.

Areas of application: Behavioural models in sociology (e.g. migration behaviour), traffic flow simulations, epidemic models.

## 3. System dynamics simulation (quasi-continuous)

Description: This simulation analyses the behaviour of complex systems over time, taking into account feedback loops and time delays. It is often used to understand large systems on a macroscopic level.

Areas of application: Management, ecosystems, urban planning.

You have now been introduced to the world of simulation. In the next and final chapter, you can test your knowledge and complete the learning unit.

# Completion of the learning unit

*Have you got the gist of it?*

At the end of the learning unit, you can check whether you have understood the essentials. Check your knowledge by answering the following 8 multiple choice questions.

If you have answered all 8 questions correctly, you will receive a badge with an award. This is visible at the top of the course. You can use it to start further self-study learning units online, which will be activated for you.

Have fun!

# Final learning questions

How does a model differ from a system?

- A system and a model are the same thing, just described using different terms.
- A model is a real replica of a system that functions in exactly the same way as the real system.
- A system is a group of parts that work together, while a model is a simplified representation of that system to better understand it. The model helps to reproduce the behaviour of the system without showing all the details.